Lab 3
นาย ภูริณัฐ บุญมาถึง

Packet Tracer - VLAN Configuration

Addressing Table

Device	Interface	IP Address	Subnet Mask	VLAN
PC1	NIC	172.17.10.21	255.255.255.0	10
PC2	NIC	172.17.20.22	255.255.255.0	20
PC3	NIC	172.17.30.23	255.255.255.0	30
PC4	NIC	172.17.10.24	255.255.255.0	10
PC5	NIC	172.17.20.25	255.255.255.0	20
PC6	NIC	172.17.30.26	255.255.255.0	30

Objectives

Part 1: Verify the Default VLAN Configuration

Part 2: Configure VLANs

Part 3: Assign VLANs to Ports

Background

VLANs are helpful in the administration of logical groups, allowing members of a group to be easily moved, changed, or added. This activity focuses on creating and naming VLANs, and assigning access ports to specific VLANs.

Part 1: View the Default VLAN Configuration

Step 1: Display the current VLANs.

On S1, issue the command that displays all VLANs configured. By default, all interfaces are assigned to VLAN 1.

Step 2: Verify connectivity between PCs on the same network.

Notice that each PC can ping the other PC that shares the same subnet.

- PC1 can ping PC4
- PC2 can ping PC5
- PC3 can ping PC6

Pings to hosts on other networks fail.

What benefits can VLANs provide to the network?

Part 2: Configure VLANs

Step 1: Create and name VLANs on S1.

a. Create the following VLANs. Names are case-sensitive and must match the requirement exactly:

- VLAN 10: Faculty/Staff

```
S1#(config)# vlan 10
S1#(config-vlan)# name Faculty/Staff
```

b. Create the remaining VLANS.

- VLAN 20: Students
- VLAN 30: Guest(Default)
- VLAN 99: Management&Native
- VLAN 150: VOICE

Step 2: Verify the VLAN configuration.

Which command will only display the VLAN name, status, and associated ports on a switch?

Step 3: Create the VLANs on S2 and S3.

Use the same commands from Step 1 to create and name the same VLANs on S2 and S3.

Step 4: Verify the VLAN configuration.

Part 3: Assign VLANs to Ports

Step 1: Assign VLANs to the active ports on S2.

a. Configure the interfaces as access ports and assign the VLANs as follows:

- VLAN 10: FastEthernet 0/11

```
S2(config)# interface f0/11
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 10
```

b. Assign the remaining ports to the appropriate VLAN.

- VLAN 20: FastEthernet 0/18
- VLAN 30: FastEthernet 0/6

Step 2: Assign VLANs to the active ports on S3.

S3 uses the same VLAN access port assignments as S2. Configure the interfaces as access ports and assign the VLANs as follows:

- VLAN 10: FastEthernet 0/11
- VLAN 20: FastEthernet 0/18
- VLAN 30: FastEthernet 0/6

Step 3: Assign the VOICE VLAN to FastEthernet 0/11 on S3.

As shown in the topology, the S3 FastEthernet 0/11 interface connects to a Cisco IP Phone and PC4. The IP phone contains an integrated three-port 10/100 switch. One port on the phone is labeled Switch and connects to F0/4. Another port on the phone is labeled PC and connects to PC4. The IP phone also has

an internal port that connects to the IP phone functions.

The S3 F0/11 interface must be configured to support user traffic to PC4 using VLAN 10 and voice traffic to the IP phone using VLAN 150. The interface must also enable QoS and trust the Class of Service (CoS) values assigned by the IP phone. IP voice traffic requires a minimum amount of throughput to support acceptable voice communication quality. This command helps the switchport to provide this minimum amount of throughput.

```
S3(config)# interface f0/11
S3(config-if)# mls qos trust cos
S3(config-if)# switchport voice vlan 150
```

Step 4: Verify loss of connectivity.

Previously, PCs that shared the same network could ping each other successfully.

Study the output of from the following command on **S2** and answer the following questions based on your knowledge of communication between VLANS. Pay close attention to the Gig0/1 port assignment.

```
S2# show vlan brief
VLAN Name Status Ports
---- ------------------------------ --------- ------------------------------
1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/7, Fa0/8, Fa0/9
Fa0/10, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/1, Gig0/2
10 Faculty/Staff active Fa0/11
20 Students active Fa0/18
30 Guest(Default) active Fa0/6
99 Management&Native active
150 VOICE active
```

Try pinging between PC1 and PC4.

Although the access ports are assigned to the appropriate VLANs, were the pings successful? Explain.

What could be done to resolve this issue?

ตอบ

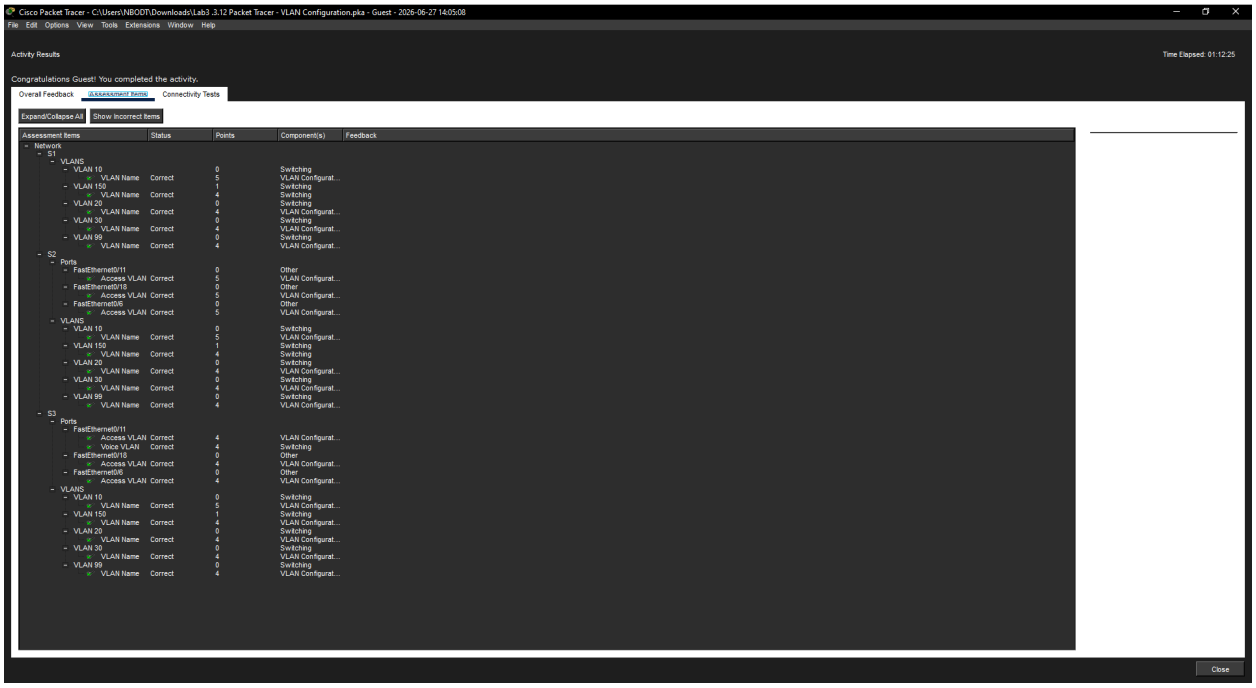
คำถาม:

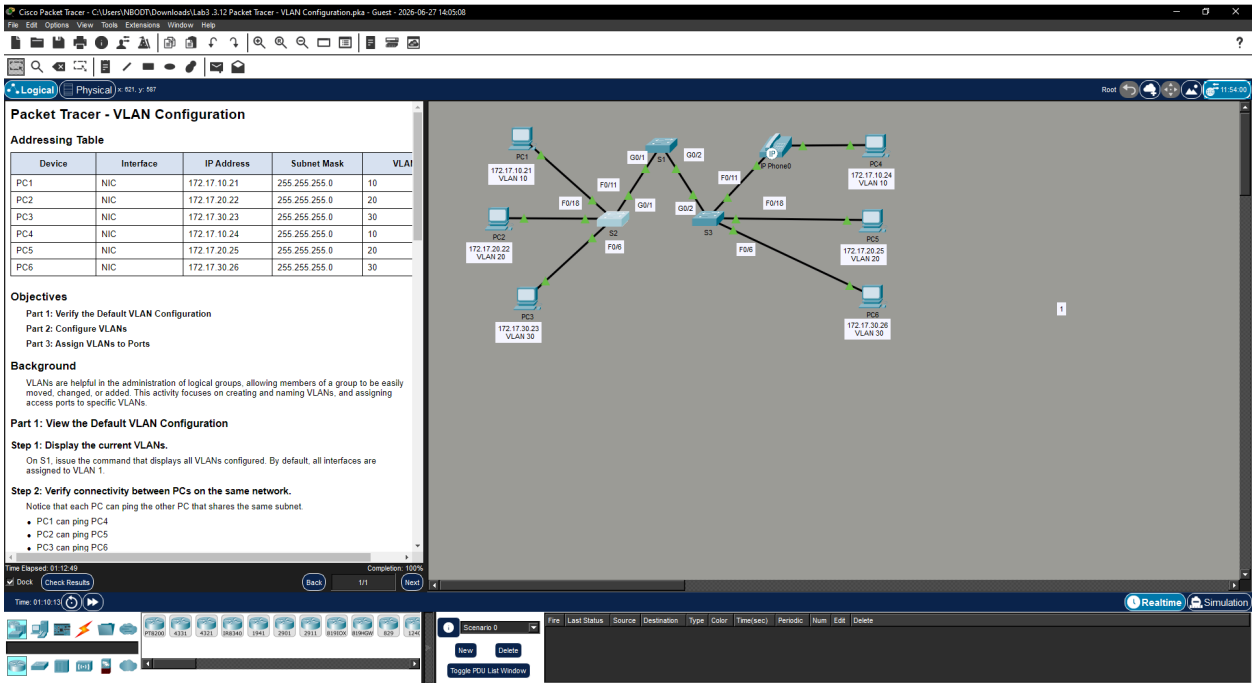
1. แม้ว่าพอร์ตใช้งาน (Access Ports) จะถูกกำหนดให้กับ VLAN ที่ถูกต้องแล้ว แต่การ Ping ประสบความสำเร็จหรือไม่? เพราะเหตุใด?

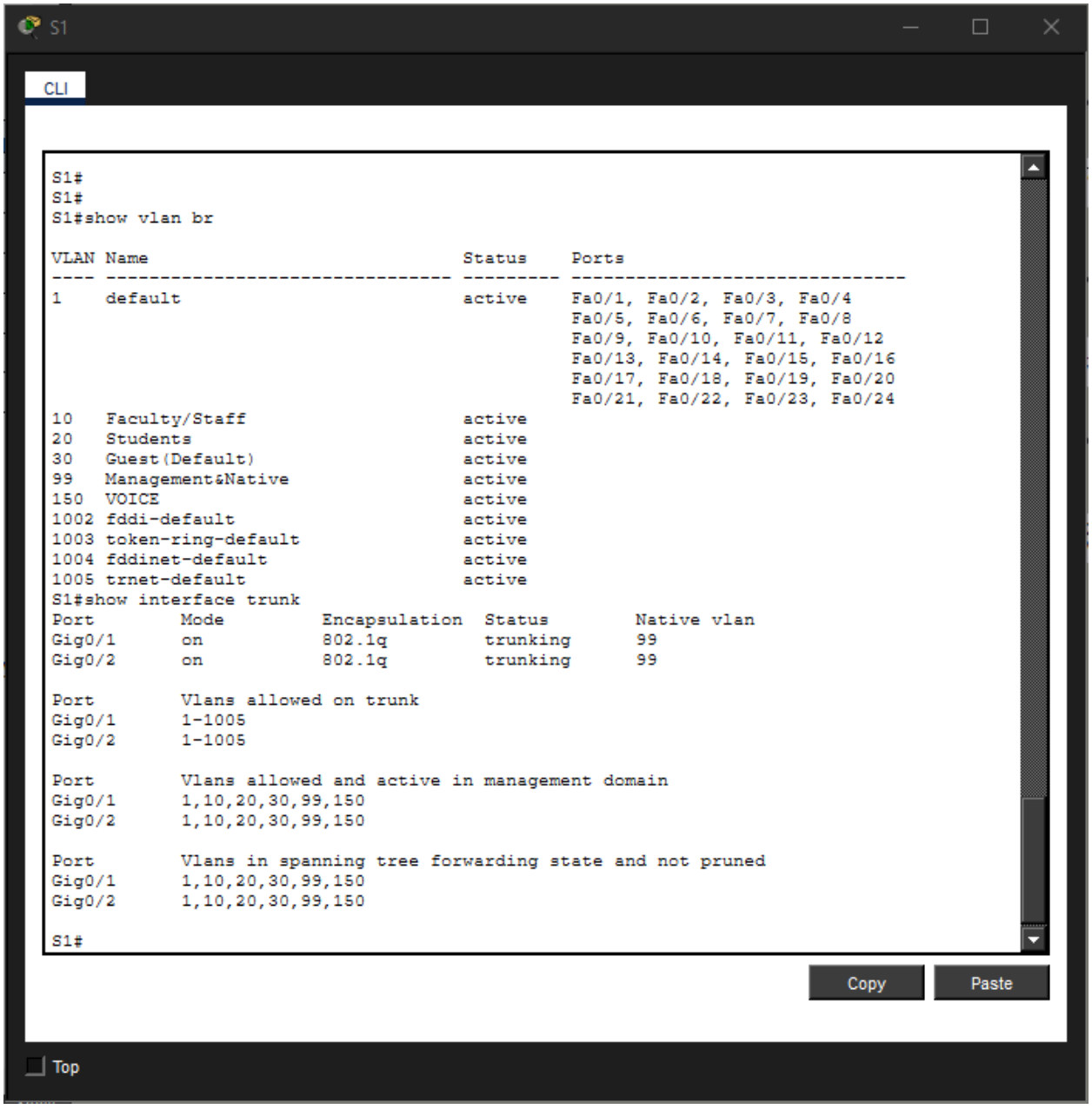
> **คำตอบ: ไม่สำเร็จ (Ping ล้มเหลว)** เนื่องจากพอร์ตเชื่อมต่อระหว่างสวิตช์ (พอร์ตอัปลิงก์ เช่น `Gig0/1` และ `Gig0/2`) ยังคงถูกจัดอยู่ใน VLAN 1 และทำงานเป็นพอร์ตเข้าถึง (Access Port) ธรรมดา ทำให้มันปฏิเสธหรือทำลายทราฟฟิก (Drop) ที่ติดแท็กของ VLAN 10, 20 หรือ 30 ที่พยายามจะวิ่งข้ามไปยังสวิตช์ตัวอื่น

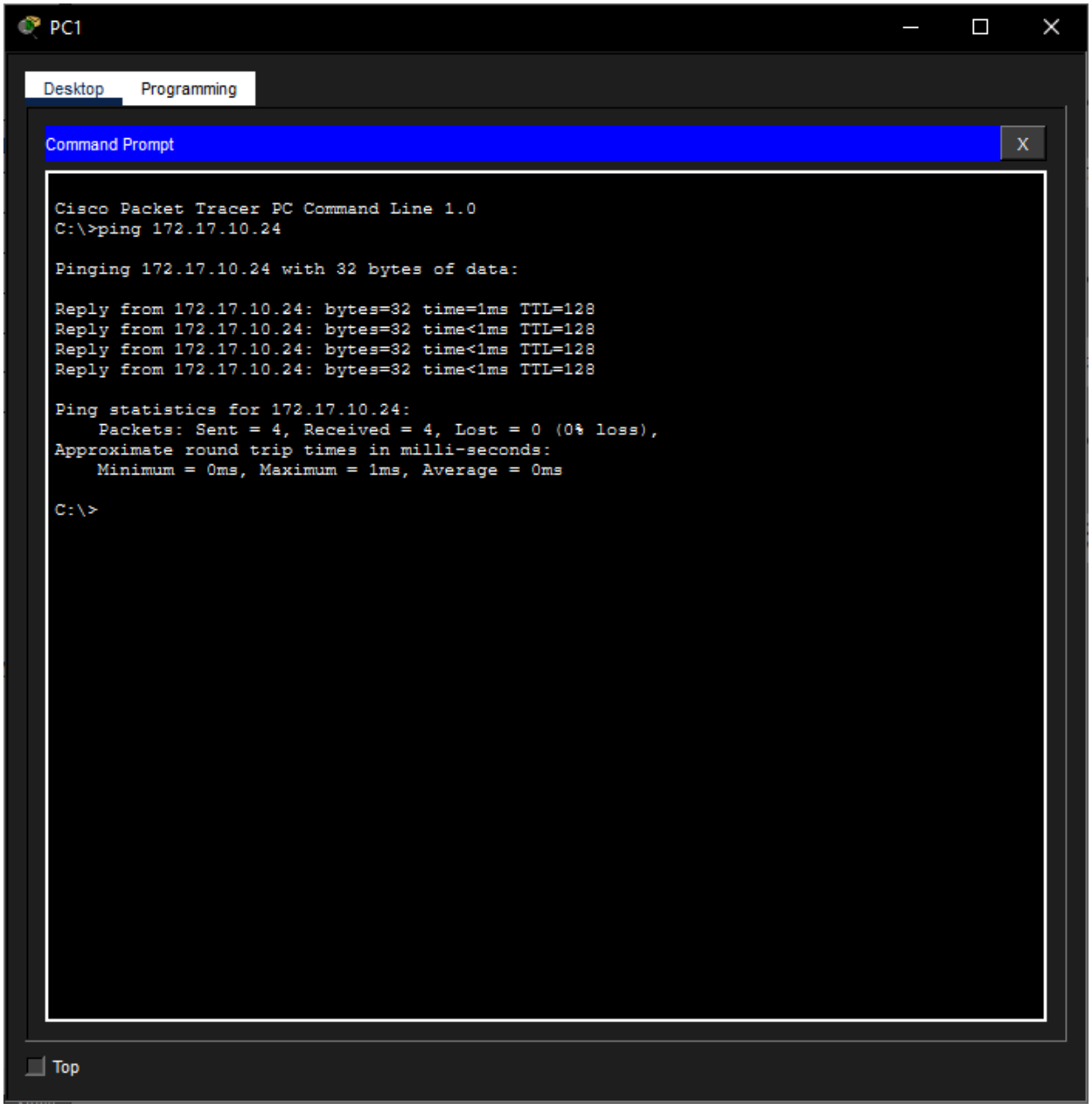
2. ควรทำอย่างไรเพื่อแก้ไขปัญหานี้?

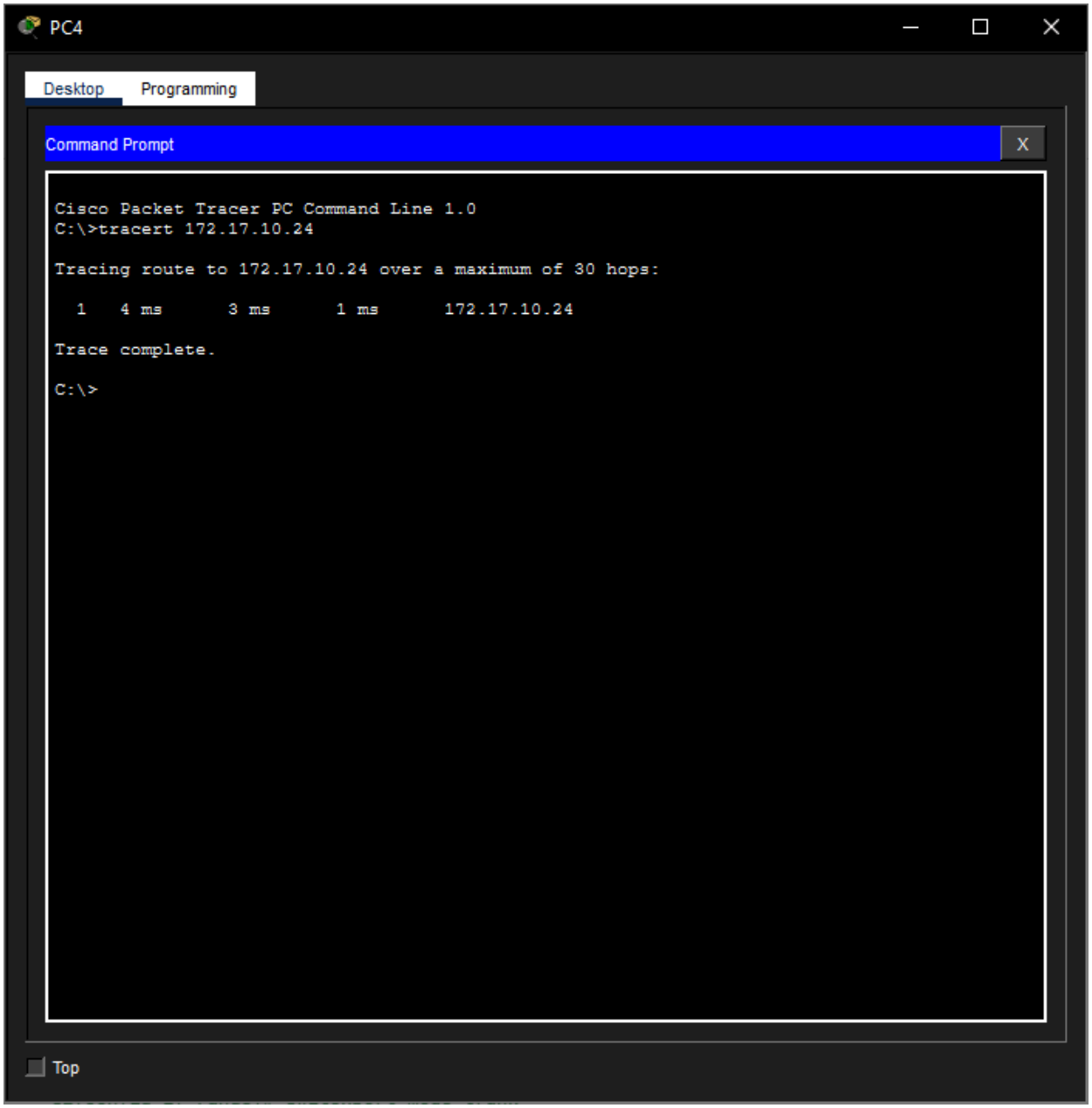
คำตอบ: ต้องเปลี่ยนโหมดการทำงานของพอร์ตเชื่อมต่อระหว่างสวิตช์ทั้งหมด (จาก S1 ไป S2 และจาก S1 ไป S3) ให้เป็น พอร์ตทรังก์ เพื่อให้มีสายสัญญาณเส้นเดียวสามารถส่งผ่านทราฟฟิกของ VLAN พร้อมกันหลายๆ VLAN และสามารถข้ามไปยังสวิตช์ตัวอื่นได้โดยใช้การแนบแท็กระบุตัวตน

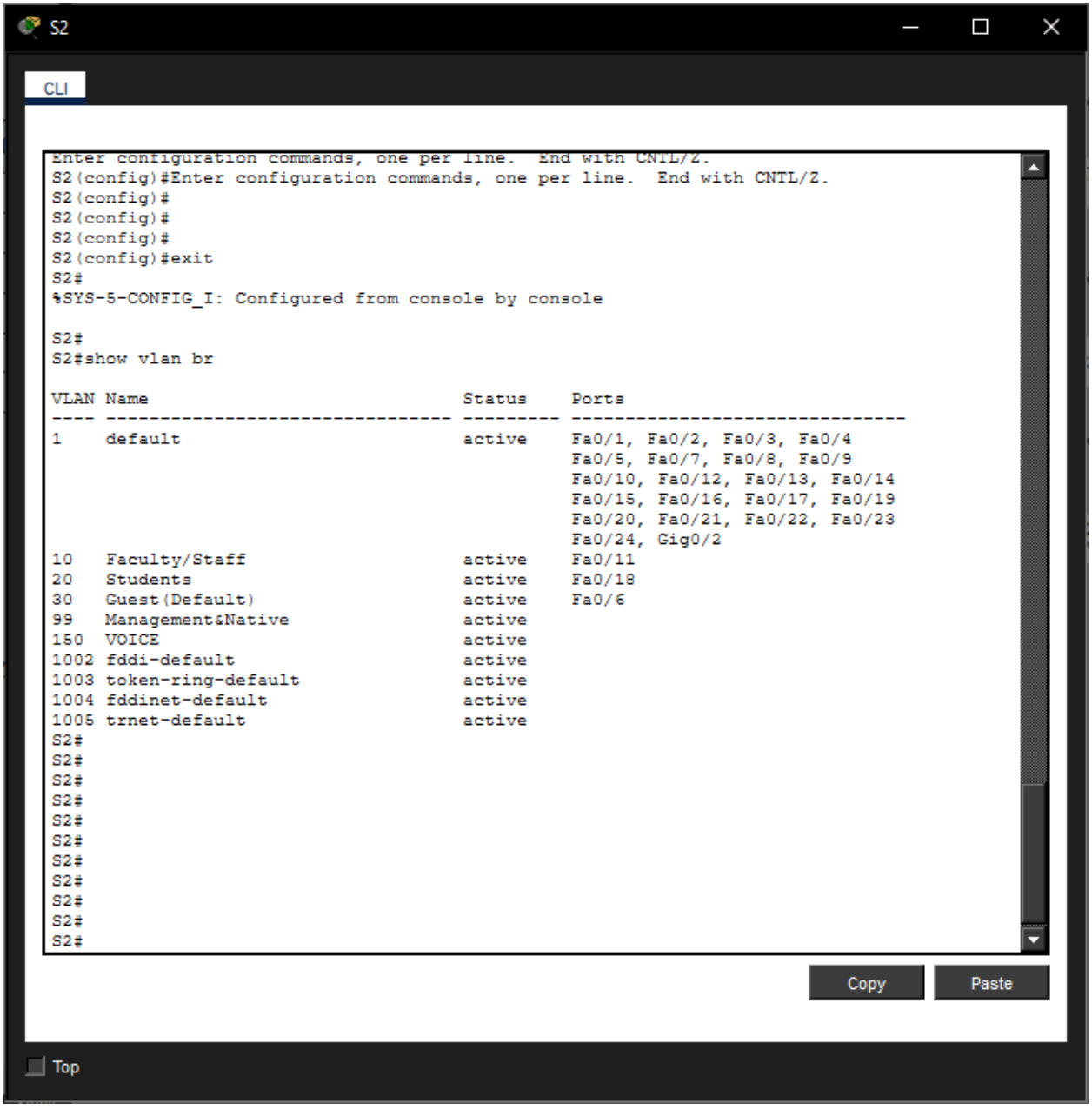


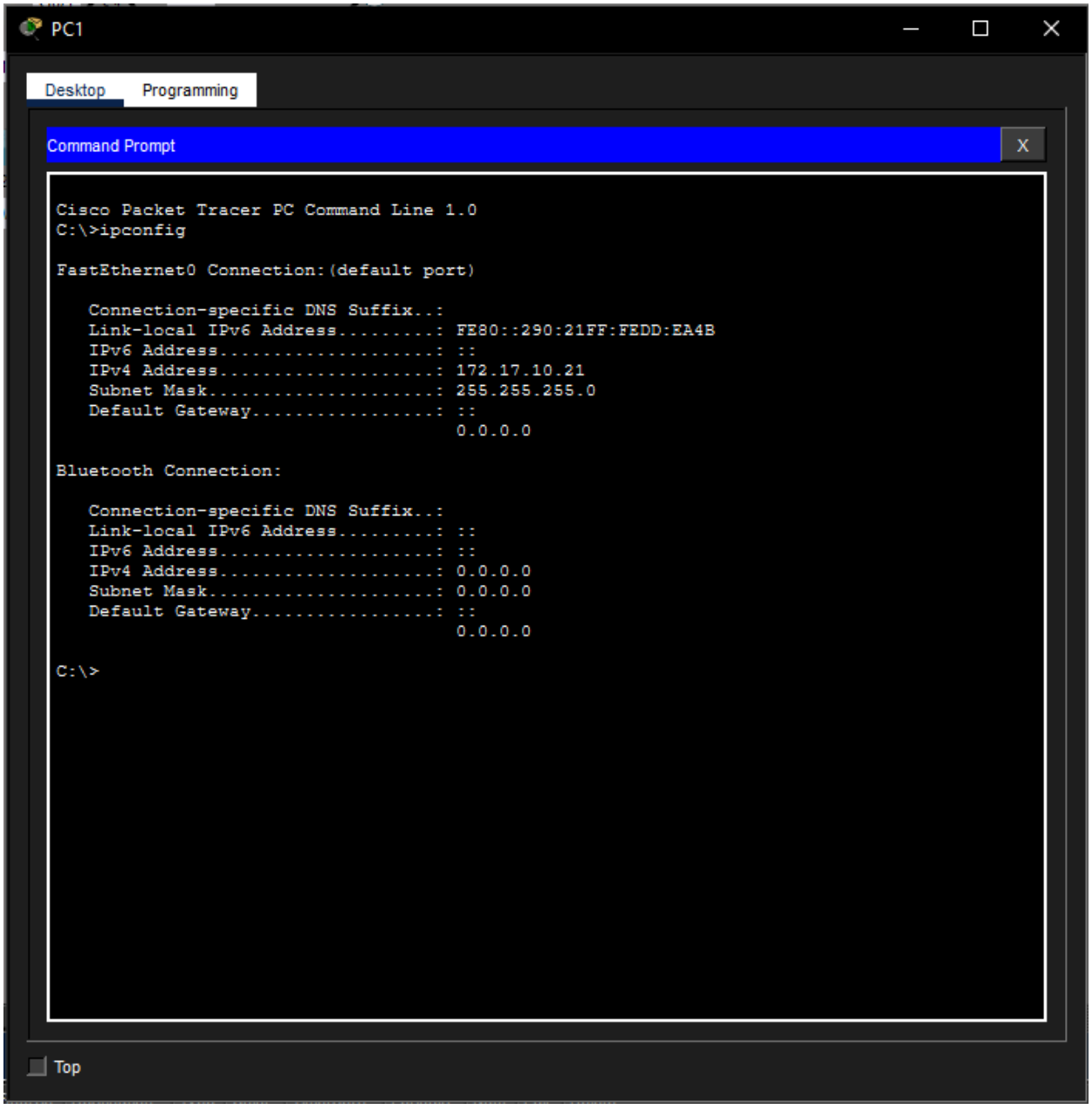



S1
CLI
S1#
S1#
S1#show vlan br
VLAN Name Status Ports
---- -------------------------------- --------- -------------------------------
1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/6, Fa0/7, Fa0/8
Fa0/9, Fa0/10, Fa0/11, Fa0/12
Fa0/13, Fa0/14, Fa0/15, Fa0/16
Fa0/17, Fa0/18, Fa0/19, Fa0/20
Fa0/21, Fa0/22, Fa0/23, Fa0/24
10 Faculty/Staff active
20 Students active
30 Guest(Default) active
99 Management&Native active
150 VOICE active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
S1#show interface trunk
Port Mode Encapsulation Status Native vlan
Gig0/1 on 802.1q trunking 99
Gig0/2 on 802.1q trunking 99
Port Vlans allowed on trunk
Gig0/1 1-1005
Gig0/2 1-1005
Port Vlans allowed and active in management domain
Gig0/1 1,10,20,30,99,150
Gig0/2 1,10,20,30,99,150
Port Vlans in spanning tree forwarding state and not pruned
Gig0/1 1,10,20,30,99,150
Gig0/2 1,10,20,30,99,150
S1#
Copy
Paste
Top


PC1
Desktop
Programming
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.17.10.24
Pinging 172.17.10.24 with 32 bytes of data:
Reply from 172.17.10.24: bytes=32 time=1ms TTL=128
Reply from 172.17.10.24: bytes=32 time<1ms TTL=128
Reply from 172.17.10.24: bytes=32 time<1ms TTL=128
Reply from 172.17.10.24: bytes=32 time<1ms TTL=128
Ping statistics for 172.17.10.24:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top


PC4
Desktop
Programming
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 172.17.10.24
Tracing route to 172.17.10.24 over a maximum of 30 hops:
1 4 ms 3 ms 1 ms 172.17.10.24
Trace complete.
C:\>
Top


S2
CLI
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#
S2(config)#
S2(config)#
S2(config)#exit
S2#
%SYS-5-CONFIG_I: Configured from console by console
S2#
S2#show vlan br
VLAN Name Status Ports
1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/7, Fa0/8, Fa0/9
Fa0/10, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/2
10 Faculty/Staff active Fa0/11
20 Students active Fa0/18
30 Guest(Default) active Fa0/6
99 Management&Native active
150 VOICE active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
S2#
S2#
S2#
S2#
S2#
S2#
S2#
S2#
S2#
S2#
S2#
S2#
Copy
Paste
Top


PC1
Desktop
Programming
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::290:21FF:FEDD:EA4B
IPv6 Address.....: ::
IPv4 Address.....: 172.17.10.21
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>
Top